

Detailed Test Design and Execution

1. Selected Major Module

This document treats `coupon_discount_engine` as the selected major feature of the target e-commerce promotion application. It is not the whole AutoTestDesign tool itself. Instead, it is the executable feature used to demonstrate that the test designs produced or refined with ARG-Test can be translated into concrete black-box and white-box tests.

This module is a strong detailed-execution target because it combines:

- valid and invalid input partitions
- multiple numeric boundaries
- interacting business rules
- explicit expected results
- a compact reference implementation suitable for white-box execution

The module is suitable for detailed execution because its expected behavior is explicit, reviewable, and rich enough to require more than one testing technique. A single nominal-path test would not be sufficient: the module has rejection rules, threshold rules, membership restrictions, and implementation branches that must be checked together.

2. Requirement Basis and Scope

2.1 Formalized Requirement Summary

The detailed execution document uses the following normalized rules derived from the final requirement set and the selected reference implementation:

- R1: only one coupon may be applied to an order
- R2: unknown coupon codes must be rejected
- R3: expired coupons must be rejected
- R4: `SAVE10` requires `subtotal >= 50`
- R5: `SAVE20` requires `subtotal >= 100`
- R6: `SAVE20` also requires premium membership
- R7: `SAVE20` cannot be combined with sale items
- R8: `FREESHIP` requires `subtotal >= 30`
- R9: no-coupon input should leave subtotal and shipping unchanged

These rules define the coverage items used in the rest of the document. The black-box design focuses on externally visible behavior from these rules, while the white-box design checks whether the executable reference implementation exercises all important branches that implement them.

2.2 Implementation Under Test

Item	Path
Reference implementation	reference_impl/coupon_discount_engine.py
Black-box tests	tests/test_coupon_discount_engine_blackbox.py
White-box tests	tests/test_coupon_discount_engine_whitebox.py
Seeded mutants	reference_impl/coupon_discount_engine_mutants.py

3. Test Environment and Tooling

Tool	Purpose
pytest	execute black-box and white-box test cases
coverage.py	collect statement and branch coverage
mutant functions	demonstrate defect detection usefulness

pytest was selected because the reference implementation is a compact Python module and the expected outcomes can be expressed directly with assertions. coverage.py was added to connect the test design with executable evidence. Branch coverage is especially useful here because coupon logic is implemented through conditionals, not only straight-line statements.

Commands used:

```
python -m pytest tests\test_coupon_discount_engine_blackbox.py tests\test_coupon_discount_engine_whitebox.py
python -m pytest tests -q
python -m coverage run --branch -m pytest tests -q
python -m coverage report -m reference_impl\coupon_discount_engine.py
python -m coverage xml -o final_docs\execution_evidence\coupon_discount_engine_coverage.xml
python -m coverage xml -o final_docs\execution_evidence\coupon_discount_engine_branch_coverage.xml
```

4. Black-Box Test Design

The black-box design was derived from the requirement rules without relying on internal implementation details. Three techniques are used together because they cover different risks. Equivalence Partitioning separates valid and invalid classes. Boundary Value Analysis targets monetary thresholds where off-by-one or inclusive/exclusive mistakes are likely. Decision Table Testing captures business-rule combinations, especially for SAVE20, where multiple conditions interact.

4.1 Equivalence Partitioning

Partition ID	Partition type	Representative condition	Designed test(s)	Expected result
EP1	valid	no coupon provided	BB01	accept and keep subtotal/shipping unchanged
EP2	invalid	more than one coupon provided	BB02	reject with one-coupon-only reason

Partition ID	Partition type	Representative condition	Designed test(s)	Expected result
EP3	invalid	unknown coupon code	BB03	reject with unknown-coupon reason
EP4	invalid	expired coupon	BB04	reject with expired-coupon reason
EP5	invalid	premium-only coupon used by non-premium customer	BB07	reject with membership reason
EP6	invalid	restricted coupon combined with sale items	BB08	reject with sale-item restriction reason
EP7	valid	SAVE20 with all preconditions satisfied	BB09	accept and apply 20% discount

4.2 Boundary Value Analysis

Boundary ID	Threshold	Below	On boundary	Above / valid representative	Designed test(s)
B1	SAVE10 subtotal 50	49	50	60	BB05, BB06, WB05
B2	SAVE20 subtotal 100	99	100	120	WB03, BB09
B3	FREESHIP subtotal 30	29	30	80 with no coupon	WB04, BB10, BB01

4.3 Decision Table

Rule	Coupon	Threshold met	Premium member	Sale items present	Expired	Expected outcome
D1	none	N/A	N/A	N/A	N/A	accept, no change
D2	SAVE10	no	N/A	N/A	no	reject
D3	SAVE10	yes	N/A	N/A	no	accept, 10% discount
D4	SAVE20	yes	no	no	no	reject
D5	SAVE20	yes	yes	yes	no	reject
D6	SAVE20	yes	yes	no	no	accept, 20% discount
D7	FREESHIP	no	N/A	N/A	no	reject
D8	FREESHIP	yes	N/A	N/A	no	accept, shipping becomes zero

4.4 Executable Black-Box Cases

Test ID	pytest function	Main technique	Covered rule / purpose
BB01	test_no_coupon_keeps_order_values	EP	no-coupon valid partition
BB02	test_multiple_coupons_are_rejected	EP	one-coupon-only rejection

Test ID	pytest function	Main technique	Covered rule / purpose
BB03	test_unknown_coupon_is_rejected	EP	unknown coupon rejection
BB04	test_expired_coupon_is_rejected	EP	expired coupon rejection
BB05	test_save10_boundary_below_threshold_is_rejected	BVA	SAVE10 just below threshold
BB06	test_save10_boundary_on_threshold_is_accepted	BVA	SAVE10 exact threshold acceptance
BB07	test_save20_requires_premium_membership	EP / decision table	premium requirement
BB08	test_save20_with_sale_items_is_rejected	EP / decision table	sale-item restriction
BB09	test_save20_valid_case_applies_discount	decision table	valid SAVE20 acceptance
BB10	test_freeship_threshold_on_boundary_sets_shipping_to_zero	BVA	FREESHIP exact threshold acceptance

4.5 Requirement-to-Test Traceability

The following traceability matrix connects the requirement rules, coverage ideas, selected techniques, executable test IDs, and expected behavior. This is the main bridge between the abstract test design and the concrete `pytest` implementation.

Requirement	Coverage item	Technique	Test ID(s)	pytest function(s)	Expected behavior
R1	more than one coupon	EP	BB02	test_multiple_coupons_are_rejected	reject multiple coupons
R2	unknown coupon code	EP	BB03	test_unknown_coupon_is_rejected	reject unknown code
R3	expired coupon	EP	BB04	test_expired_coupon_is_rejected	reject expired coupon
R4	SAVE10 below/on threshold	BVA	BB05, BB06, WB05	test_save10_boundary_below_threshold_is_rejected, test_save10_boundary_on_threshold_is_accepted, test_coupon_normalization_accepts_mixed_case_and_spacing	reject below 50, accept at 50 or above
R5	SAVE20 threshold	BVA / decision table	WB03, BB09	test_save20_below_threshold_keeps_original_values, test_save20_valid_case_applies_discount	reject below 100, accept when all conditions hold

Requirement	Coverage item	Technique	Test ID(s)	pytest function(s)	Expected behavior
R6	premium membership required	EP / decision table	BB07	test_save20_requires_premium_membership	reject non-premium customer
R7	sale-item restriction	EP / decision table	BB08	test_save20_with_sale_items_is_rejected	reject SAVE20 with sale items
R8	FREESHIP threshold	BVA	WB04, BB10	test_freeship_below_threshold_is_rejected, test_freeship_threshold_on_boundary_sets_shipping_to_zero	reject below 30, set shipping to zero at 30
R9	no coupon path	EP	BB01	test_no_coupon_keeps_order_values	accept and keep values unchanged

5. White-Box Test Design

5.1 White-Box Objectives

The white-box design complements the black-box suite by checking implementation branches that may not be fully justified by a single external rule. For example, the negative value guards are implementation-level safety checks, while coupon normalization verifies that the function handles user-facing input variations such as whitespace and mixed letter case.

The white-box design targets:

- input guard branches for negative values
- all coupon dispatch branches
- rejection branches for each invalid condition
- acceptance branches for SAVE10, SAVE20, and FREESHIP
- normalization behavior for mixed-case and whitespace coupon input

5.2 Branch-to-Test Mapping

White-box obligation	Designed test(s)
subtotal < 0 guard raises ValueError	WB01
shipping_fee < 0 guard raises ValueError	WB02
SAVE20 threshold-reject branch	WB03
FREESHIP threshold-reject branch	WB04
coupon normalization branch	WB05
SAVE10 accept path	BB06, WB05
SAVE20 accept path	BB09
FREESHIP accept path	BB10

5.3 Executable White-Box Cases

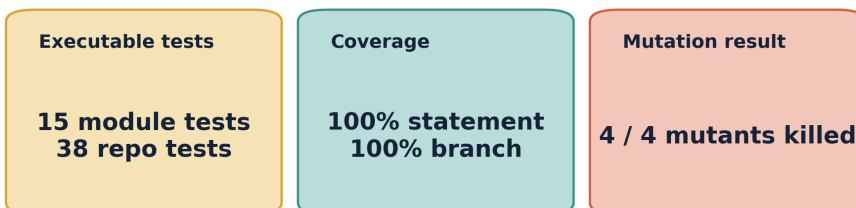
Test ID	pytest function	Covered white-box purpose
WB01	test_negative_subtotal_raises_value_error	negative subtotal guard
WB02	test_negative_shipping_fee_raises_value_error	negative shipping fee guard
WB03	test_save20_below_threshold_keeps_original_values	SAVE20 threshold rejection branch
WB04	test_freeship_below_threshold_is_rejected	FREESHIP rejection branch
WB05	test_coupon_normalization_accepts_mixed_case_and_spacing	normalization and SAVE10 valid path

Together, the black-box and white-box tests form a layered design. The black-box part proves that the observable business rules are represented. The white-box part proves that the implementation paths behind those rules are executed, including defensive branches and normalization behavior.

6. Execution Results

Detailed module execution evidence

coupon_discount_engine combines black-box design, white-box execution, and mutation usefulness evidence.



Evidence sources: tests/test_coupon_discount_engine_blackbox.py, tests/test_coupon_discount_engine_whitebox.py, coverage XML, and coupon_discount_engine_mutation_demo.md.

Detailed Module Execution Evidence

6.1 Summary of Observed Results

Item	Observed result
Module-specific executable tests	15 passed
Repository regression suite at report-preparation time	38 passed
Statement coverage on the reference module	100%
Branch coverage on the reference module	100%
Mutation result	4 / 4 mutants killed

6.2 Coverage Summary

Name	Stmts	Miss	Branch	BrPart	Cover
reference_impl\coupon_discount_engine.py	51	0	26	0	100%
TOTAL	51	0	26	0	100%

Coverage interpretation:

- black-box design is strong enough to exercise the functional rule structure
- white-box design closes the remaining branch obligations
- no branch in the selected reference implementation remains unexecuted

6.3 Result Analysis

The detailed execution result is strong for three reasons.

First, the black-box suite is not superficial. It covers valid partitions, invalid partitions, multiple thresholds, and interacting rule conditions such as premium membership plus sale-item restrictions.

Second, the white-box suite is not decorative. It exercises the explicit negative-input guards and rejection/acceptance branches that are easy to overlook in a purely requirement-level discussion.

Third, the combined suite is compact. With only 15 module-focused cases, it achieves complete statement and branch coverage on the selected implementation, which is a good tradeoff between completeness and maintainability for this module-level validation.

7. Web Demo Consistency Validation

The final delivery also validates that the Web demo presents the same formal evidence used by the report instead of mixing official results with unrelated local mock outputs. This check matters because the demo is the primary presentation surface for the final defense, while the formal result bundle remains the source of quantitative evidence.

SOFTWARE TESTING FINAL PROJECT

ARG-Test Demo Console

Requirement-driven test design with stable mock interaction and frozen formal evidence.

Direct Input

CSV Batch

State Model

Formal Evidence

RECORDED-DEMO POLICY

Mock for interaction. Frozen run for claims.

Show the app flow, then cite the formal dashboard for final quality numbers.

LIVE INTERACTION

Direct Requirement Input

Choose a frozen test requirement; the official ID, split, and requirement text are filled automatically.

Requirement Case

Split

pickup_station_contact_validation

test

Provider

Model

mock

mock-arg-test

Candidates

Seed

3

202601

Auto-filled Requirement Text

Requirement ID: pickup_station_contact_validation

Domain: pickup fulfillment validation

Rules:

1. Required fields: recipient_name, recipient_phone, pickup_station_id.

2. pickup_station_id must match PS- followed by 6 digits.

3. recipient_phone must be in E.164 format (+ followed by 8 to 15 digits).

4. pickup_code is optional unless contactless_pickup is true.

5. When contactless_pickup is true, pickup_code must be exactly 6 alphanumeric characters.

Choose a frozen test requirement to auto-fill its text. Keep provider as mock for stable recorded interaction.

Generate Test Suite

Completed.

OUTPUT PREVIEW

Generated Suite and Diagnostics

STRUCTURAL CHECKER

0.950

Contract checks; category: input_validation

OVERALL COVERAGE

71.3%

9 frozen test cases

RUN MODE

Replay

Frozen formal output

REQUIREMENT

pickup_station_contact_validation

test

Result Source

Frozen formal replay

Matches Formal Evidence coverage

No live API call

Risk Assessment

RISK LEVEL

Medium

Priority signal for generated cases

RISK SCORE

6.700

7 rules, 2 numeric constraints

Drivers

category=input_validation

7 explicit rule lines

2 numeric constraints

decision-table logic

strict rejection/constraint language: required, unless

Recommended Focus

core happy path

rule combination coverage

boundary coverage

decision combinations

invalid partitions

negative/error handling

Diagnostics

demo_mode: frozen formal replay; coverage matches the official Formal Evidence dashboard

bva_contract: missing on-boundary case

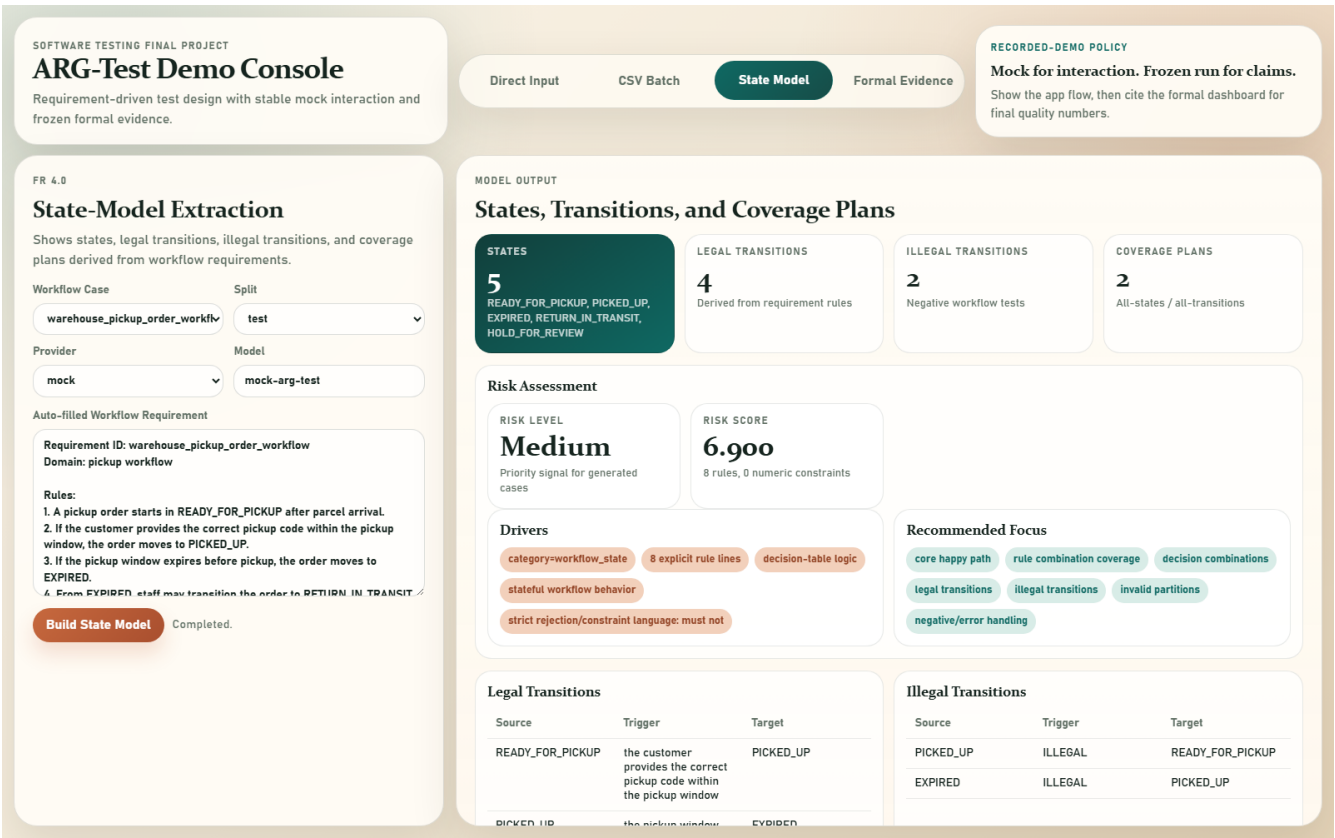
Direct Frozen Replay Evidence

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The Direct tab now uses a formal requirement catalog. When a built-in formal test requirement is selected, the backend returns frozen_formal_run replay data from the archived result snapshot. This prevents mismatches such as a selected label showing one coverage value while the generated local mock result displays another value.

Web demo consistency check	Observed result
Direct-tab formal catalog replay	16 / 16 formal test requirements replayed frozen outputs
CSV sample replay	2 / 2 matching CSV rows replayed frozen outputs
Formal dashboard aggregate	Average overall coverage shown as 61.5%
Repository regression after Web demo fixes	38 passed

The State-Model tab was also checked on the five workflow examples included in the demo catalog. Each catalog workflow now produces a non-empty legal-transition model, and illegal transitions are only reported when the requirement text explicitly supports them.



State Model Extraction Evidence

Workflow example	States	Legal transitions	Illegal transitions
order_approval_state_machine	6	7	0
order_split_shipment_state_machine	4	3	2
payment_3ds_authentication_flow	6	6	2
return_exchange_approval_workflow	9	8	1
warehouse_pickup_order_workflow	5	4	2

This Web-level validation does not replace the detailed executable module evidence. Instead, it protects the final demonstration from interpretation errors: official quantitative results come from frozen formal replay, while ad hoc edited inputs remain clearly labeled as local mock generation.

8. Mutation-Based Usefulness Demonstration

The evaluation also includes defect-seeded usefulness evidence. Four representative mutants were created:

- a mutant that allows multiple coupons
- a mutant that rejects SAVE10 exactly at subtotal 50
- a mutant that ignores the SAVE20 sale-item restriction
- a mutant that rejects FREESHIP exactly at subtotal 30

Observed outcome:

- 4 / 4 mutants were killed

Mutation-to-test mapping:

Mutant	Killed by
multiple_coupons_allowed	BB02, BB08
save10_boundary_bug	BB06
save20_sale_item_bug	BB08
freeship_boundary_bug	BB10

This is important because it shows that the detailed suite is not only structurally complete. It is also effective at detecting realistic logic defects that correspond to the selected business rules and boundaries.

9. Evidence Paths

Primary evidence files:

- final_docs/execution_evidence/coupon_discount_engine_execution_summary.md
- final_docs/execution_evidence/coupon_discount_engine_coverage.xml
- final_docs/execution_evidence/coupon_discount_engine_branch_coverage.xml
- final_docs/execution_evidence/coupon_discount_engine_mutation_demo.md
- demo_web/app.py
- tests/test_demo_web_api.py
- report_assets/final_demo_package/frontend_focus/formal_results_snapshot/

10. Conclusion

The `coupon_discount_engine` module is supported by a complete detailed-design and execution chain. The module is covered by multiple black-box techniques, executable white-box tests, complete statement and branch coverage, and a successful mutation demonstration. This makes it a credible detailed anchor for the overall ARG-Test submission rather than a merely illustrative example.

Most importantly, the evidence chain is complete: requirement rules are transformed into coverage items, coverage items are mapped to executable `pytest` tests, the tests are run against a reference implementation, and the resulting suite is checked with both coverage and mutation evidence. The result is a traceable module-level validation package that connects test case design, test tool implementation, and test result analysis.